

SENTINEL-AI: FORENSIC INTELLIGENCE REPORT

Lead Analyst: Shalom Kumar | Session: 2025-26 | Version: 4.5.0

Institution: ISBM University | Faculty of Computer Applications

Advanced Pattern Recognition & Heuristic Weightage Security Audit

SECTION 1: Target Ingestion & Identification

Target Telemetry Stream: data:text/html;base64,PHNjcmlwdD5hbGVydCgnSGFja2VkYjYpOzwvc2NyaXB0Pg==

Forensic Audit ID: SENT-DCDDDB6A0

Analysis Timestamp: 2026-05-18 | 19:31:31

SECTION 2: Intelligence-Based Risk Metrics

Heuristic Risk Index: 45/100

Security Verdict: SUSPICIOUS

SECTION 3: Detailed Forensic Logic Steps

The Sentinel-AI engine executed a weighted heuristic audit. The following threat signatures and behavioral anomalies were identified within the telemetry packet:

[!] DETECTION_NODE_1: OBFUSCATION SIGNATURE DETECTED

SECTION 3.5: Statutory Cyber Law Mapping Analysis Matrix

STEP 1: PERIMETER INGESTION & VOLUMETRIC TRAFFIC MONITORING

Technical Context Captured: IP Gateway Evaluation - Volumetric Traffic Clearance Passed [Weight: 15%]

LAW COMPLIANCE BOUNDARY: Section 43(f) of the Information Technology (IT) Act, 2000

Statutory Provision: Prevention of Denial of Service (DoS) and Unauthorized System Resource Depletion.

Forensic Verification Audit: The boundary subsystem successfully calculated the incoming transaction frequency profile of the packet stream. Rate limiting thresholds successfully prevented infrastructure thread starvation, ensuring network baseline availability.

STEP 2: HEURISTIC PATTERN SCRAMBLING & DEEP SYNTAX ANALYSIS

Technical Context Captured: RegEx Logic Node Sweep Over Telemetry Token Components [Weight: 45%]

LAW COMPLIANCE BOUNDARY: Section 66 Read With Section 43(i) of the IT Act, 2000

Statutory Provision: Computer Related Offences - Tampering, Destroying, or Altering Destructive Computer Source Code Assets.

Forensic Verification Audit: Heuristic inspection matrix executed tokenized pattern evaluation loops. Parallel processing nodes scanned data string configurations for unauthorized system administration calls (eval/exec commands), neutralizing destructive source anomalies.

STEP 3: NETWORK TRAJECTORY TRACING & REDIRECT LOOP ISOLATION

Technical Context Captured: Layer 3 Sniffer Loop Interception - Trajectory Routing Status [Weight: 0%]

LAW COMPLIANCE BOUNDARY: Section 43(f) & Section 66F of the Information Technology Act, 2000

Statutory Provision: Protection Against Cyber Terrorism and Attempted Infrastructure Lockout via Resource Exhaustion Loops.

Forensic Verification Audit: Active look-ahead sniffer framework intercepted remote host response headers. Redirection ceiling constraints successfully killed recursive redirect targets (Hmax Limit: 3) to enforce system self-preservation.

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STEP 4: CRYPTOGRAPHIC EVIDENCE SEALING & IMMUTABILITY RECONCILIATION

Technical Context Captured: **SHA-256 System Non-Repudiation Structural Signature Block [Weight: 45%]**

LAW COMPLIANCE BOUNDARY: Section 65B of Indian Evidence Act & Section 66C of IT Act, 2000

Statutory Provision: Admissibility of Electronic Records and Protection of Identity Integrity Verification Metrics.

Forensic Verification Audit: System combined raw telemetry inputs, server session keys, and risk indicators into an immutable string hash block. Localized SHA-256 calculation sealed the digital evidence structure with a unique 64-character alphanumeric signature code.

SECTION 4: Forensic Integrity Verification

SHA-256 DIGITAL FINGERPRINT (NON-REPUDIATION):

f1f36721b1c696770f8e43de4fc12013f2cee3fcd88aa67ecc4aa80a8910fef6

EXECUTIVE SUMMARY: Sentinel-AI has concluded the audit for the specified telemetry. The resulting 45% risk index warrants **ROUTINE MONITORING**. Data persisted to RDBMS logs under cryptographic law-compliance verification frameworks.